QUANT

Time, Speed & Distance

Assignment 01

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Time, Speed & Distance Online Assignment -01

1. I travel the first hour at an average speed of 50 km/h and the second hour at an average speed of 60 km/h. What was the average speed (in kmph) for the journey?

1. 55 2. 600/11 3. 58 4. 54

2. A thief flees city A in a car towards city B on a stretch of straight road, 300 kilometre long, at a speed of 60 km/hr. In 15 minutes a police party (X) leaves city A to chase the thief at 65 km/hr. After leaving, how long does it take for the police party X to catch the thief?

1. 2 hrs 45 min 2. 3 hrs 3. 3 hrs 15 min 4. None of these

3. There is a road that connects two cities A and B with a distance fo 20 km in them. Along the road there are three gutters 1, 2 and 3. The distance of city A from gutter 1 is the same as city B from gutter 3. Also the distance between gutters 2 and 3 is twice the distance between gutter 1 and 2. A man is found injured in an accident near gutter 3. If he is not given medical treatment in 40 minutes he will die. An ambulance leaves from city A and travelling at 30 km/h, reaches gutter 1 in 5 minutes. After this it doubles its speed. It takes 1 minute for the patient to be loaded and unloaded into the ambulance. How much time will be available for the doctors to treat the patient when he arrives at the hospital?

1. 4 minutes 2. 2.5 minutes
3. 1.5 minutes 4. The patient dies before reaching the hospital

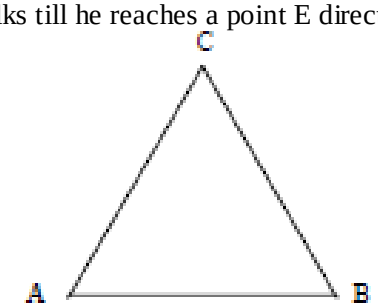
4. If I travelled at $^3/_4{}^{th}$ of my average speed and reached 25 minutes late, what is the time that I usually take to reach my destination?

1. 1 hr 2. 1.5 hr 3. 1.25 hrs 4. 1.15 hr

5. Two trains are travelling in opposite directions at uniform speeds of 60 and 50 km per hour respectively. They take 5 seconds to cross each other. If the two trains had travelled in the same direction, then a passenger sitting in the faster moving train would have overtaken the other train in 18 seconds. The lengths of the trains, in metres, are:

1. 101.78, 40 2. 101.78,50 3. 102.78,50 4. 101.78, 55

DIRECTIONS for Questions 6 and 7: ABC form an equilateral triangle in which B is 2 km from A. A person starts walking from B in a direction parallel to AC and stops when he reaches a point D directly east of C. He then reverses direction and walks till he reaches a point E directly south of C.



6. Then D is

1. 3 km east and 1 km north of A 2. 3 km east and √3 km north of A
3. √3 km east and 1 km south of A 4. √3 km west and 3 km north of A

7. The total distance walked by the person is

1. 3 km 2. 4 km 3. 2 √3 km 4. 6 km

8. The average speed of a train without stoppages is 48 kmph and average speed with stoppages is 40 kmph. How many minutes in an hour the train stops on an average?

1. 15min 2. 12 min 3. 20 min 4. 10 min

9. A train approaches a tunnel AB. Inside the tunnel a cat is located at a point 3/8th of the distance AB, measured from the entrance A. When the train whistles, the cat runs. If the cat moves to the entrance of the tunnel A, the train catches the cat exactly at the entrance. If the cat moves towards the exit B, the train catches the cat exactly at the exit. The speed of the train is greater than the cat by what order?

1. 3 ; 01 2. 4 : 01 3. 5: 01 4. None of these

10. Of the total of 250 km, I travel the first 125 km at a speed of 50 km/h and the balance at a speed of 60 km/h. What was the average speed for the journey?

11. 55 2. 600/11 3. 58 4. 54

Answer key & Explanations

Q No.	Key	Explanation
1.	1	55 km/h. Average speed, when time is the same, is simply the average of speeds.
2.	2	In 15 minutes, the thief has covered 15 kms. The relative speed between the police party and the thief is (65 – 60) = 5 km / hr. So in order to cover up the 15 km which is the lead that the thief has got, the police will take 15/5 = 3 hours. Hence 2nd Option.
3.	3	The distance of the first gutter from A is 2.5 km (also that of gutter 3 from B). Using this we can find distance between gutters 1 and 3 as 20 - (2.5 + 2.5) = 15 km. Now time taken to reach gutter 1 is 5 min. To travel the balance 15 + 17.5 = 32.5 km, it will take 32.5 minutes. It will take 1 min for loading and unloading. So total time is 5 + 32.5 + 1 = 38.5. So 1.5 minute is balance for the doctor to save the patient. Hence answer is option 3.
4.	3	Let my average speed be x kmph and let t hrs be the time taken to make the trip. So distance d = xt. This is also equal to (3x /4) (t + 25/60). Equating the two and solving for t we get t = 1.25 hours. So answer is 1.25 hours.
5.	3	The length of the slower train is 10 × 5/18 × 18 = 50 m ∴ length of the faster train is (110 × 5/18 × 5) – 50 = 102.78 m. Hence answer is option 3.
6.	2	D is in line with C which is at a vertical distance of (√3 / 2) × 2 = √3 km from A. Also D will be at a horizontal distance of (2 / 2) = 1 km from B or (2 + 1) = 3 km from A ∴ Answer (2).
7.	4	The person will walk distances BD + DB + BE = 2 + 2 + 2 = 6 km. Hence answer (4).
8.	4	If the train were to travel without stoppages, it would cover 48 km. With stoppages, the average speed reduces by (48 – 40) = 8 km ∴ (8/48) × 60 minutes = 10 minutes would be the time per hour the train stops on an average.
9.	1	Let the dist of the train from the tunnel be x and the length of the tunnel be y. Ratio of speeds in case 1 is x : 3/8 y. In the second case it is x + y : 5/8 y. Equating the two we get the ratio x : y as 3 : 2. And the ratio of the speeds = x : 3/8 y = 4 : 1. But the question is asking about greater, now the speed is greater in the ratio of 3 : 1 Hence answer is 1st option.
10.	2	Average speed, when distance is equal is = 2ab / (a + b). = 2 × 50 × 60 / (50 + 60) = 100 × 60 / 110 = 600 / 11 = 54 6/11 kmph